INCH-POUND

MIL-O-52724B(ME)
10 January 1990
SUPERSEDING
MIL-O-52724A(ME)
1 May 1980

MILITARY SPECIFICATION

OILERS, HAND, PLASTIC

This specification is approved for use within the USA Belvoir Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers plastic containers with a spout for dispensing lubricating oils under pressure.

1.2 Classification. The oilers shall be of the following sizes (see 6.2):

Size 1 - 8-ounce capacity.
Size 2 - 16-ounce capacity.

Style A - Straight Spout
Style B - Angle Spout

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

L-P-390	- Plastic, Molding and Extrusion Material, Polyethylene and Copolymers (Low, Medium and High Density).
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.

MILITARY

MIL-L-2104	- Lubricating Oil, Internal Combustion Engine, Tactical Service.
MIL-L-46167	- Lubricating Oil, Internal Combustion Engine, Arctic.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification for Marking of US Military Property.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN; NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

SOCIETY OF THE PLASTICS INDUSTRY (SPI)

SPI Technical Bulletin SP-410 - Dimensions for Plastic Bottles.

(Application for copies should be addressed to the Society of the Plastics Industry, 355 Lexington Avenue, New York, NY 10017.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Description. The oilers shall consist of a translucent polyethylene body with a removable straight or angle spout with a removable cap. The oiler shall be operated by hand pressure and shall eject oil as specified herein at temperatures specified herein.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.3 Materials. Materials shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Material deterioration prevention and control. The oilers shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the oilers may be exposed.

3.3.2 Recycled, reclaimed, and virgin materials. For the purpose of this requirement, recycled, reclaimed materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the oilers may be newly fabricated from recovered materials to the maximum extent practicable, provided the oilers produced meet all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the oilers.

3.4 Construction. The oilers with a straight or angle spout and spout cap shall be translucent polyethylene conforming to L-P-390, type 1, class L, grade 2. The oiler may be cylindrical or oval, shall have a flat bottom, and shall be of one-piece seamless construction.

3.5 Graduations. The oiler shall be graduated in increments of 2 ounces.

3.6 Threads. All threaded parts shall conform to the SPI Technical Bulletin SP-410.

3.7 Styles.

3.7.1 Style A. The style A straight spout shall be one-piece molded construction of the same material as the container. One end shall be threaded for not less than one complete thread engagement with the container, and the spout end shall be capable of securing the spout cap. The length of the straight spout shall be not more than 1.5 inches, and the opening of the smallest diameter shall be 0.3125 inch nominal. The straight spout with spout cap properly attached shall not leak, crack, or become separated when tested as specified in 4.5.2.

3.7.2 Style B. The style B angle spout shall be one-piece molded construction of the same material as the container. One end shall be threaded for not less than one complete thread engagement with the container and the spout end shall be capable of securing the spout cap. The length of the spout shall be not more

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than 3 inches, and the opening at the smallest diameter shall be .03125-inch nominal. The spout shall be bent 30 degrees from the longitudinal axis with the bend starting at the middle of the spout. The spout with spout cap properly attached shall not leak, crack, or become separated when tested as specified in 4.5.2.

3.7.3 Spout cap. The spout cap shall be reusable, and one-piece molded construction; it shall be capable of being attached to the spout and shall not crack or separate from the spout or allow the spout to leak when tested as specified in 4.5.2. The cap shall be of the captive type.

3.8 Performance. The oiler shall be filled with oil conforming to MIL-L-46167. With the spout cap removed, and with pressure applied by squeezing the oiler by hand, the oiler shall emit a steady flow of oil when in a horizontal position at temperature range of -25 °F to 125 °F. The oiler shall dispense 75 percent of the contents. The remaining 25 percent shall be dispensed with the oiler spout in a downward position, when tested as specified in 4.5.2.3.

3.9 Durability. The oiler shall not crack, break or leak when tested as specified in 4.5.2.2.

3.10 Chemical resistance. The oiler shall not crack, craze, darken, or show signs of chemical corrosion when tested as specified in 4.5.2.

3.11 Sizes.

3.11.1 Size 1. The size 1 oiler shall have a capacity of 8 ounces with a threaded neck conforming to SPI SP-410, size 24 straight or angle spout. The oiler shall be not more than 3.5 inches in width when constructed in an oval shape, or 2.5 inches in diameter when constructed in a cylindrical shape; the height, including the spout, shall be not more than 7 inches.

3.11.2 Size 2. The size 2 oiler shall have a capacity of 16 ounces with a threaded neck conforming to SPI SP-410, size 28 threaded straight or angle spout. The oiler shall be not more than 4.25 inches in width when constructed in an oval shape, or 3.25 inches in diameter when constructed in a cylindrical shape; the height, including the spout, shall be not more than 10 inches.

3.12 Identification marking. Identification of the oiler shall conform to MIL-STD-130 and shall include the specification part number (see 6.4).

3.13 Workmanship.

3.13.1 Finish. The finish of the oilers shall be smooth, and the surface shall be free from ridges and pinch-off flash, flow marks, or depressions.

3.13.2 Freedom from defects. The oilers shall be free from bubbles, cracks, pinholes, dirt, foreign matter, warpage, chipped edges, blisters, scratches, and other defects which could affect serviceability or appearance.

3.13.3 Uniformity. The oilers shall be uniform in color, texture, finish and physical properties.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor is responsible for ensuring that components and materials used are manufactured, examined, and tested as specified in referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The first article oilers shall be examined as specified in 4.5.1. Presence of one or more defect shall be cause for rejection.

4.3.2 Tests. The first article oilers shall be tested as specified in 4.5.2.1 through 4.5.2.3. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sample size shall be determined by using MIL-STD-105, table 1 and 2. A lot shall be accepted when no defects are found and rejected when one or more defects are found.

4.4.2 Examination. Samples selected as specified in 4.4.1 shall be examined for the defects specified in 4.5.1.

4.4.3 Tests. Samples selected as specified in 4.4.1 shall be tested as specified in 4.5.2.1 through 4.5.2.3.

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4.5 Inspection procedure.

4.5.1 Examination. The oilers shall be examined as specified herein for the following defects:

101. Material not as specified (see 3.3, 3.4).
102. Oilers not one-piece seamless construction or bottom not flat (see 3.4).
103. Graduations not as specified (see 3.5).
104. Threads of oiler neck and spout not as specified (see 3.6).
105. Threads distorted or containing sharp edges (see 3.6).
106. Straight spout not as specified (see 3.7.1).
107. Angle spout not as specified (see 3.7.2).
108. Spout cap missing or not as specified (see 3.7.3).
109. Dimensions not as specified for the applicable size (see 3.11).
110. Identification marking missing, incomplete, or illegible (see 3.12).
111. Part numbering system missing or incomplete (see 3.12).
112. Workmanship not as specified (see 3.13).

4.5.2 Tests.

4.5.2.1 Leakage. The oiler shall be filled with Grade 10 oil conforming to MIL-L-2104, completely assembled, and positioned with the spout facing downward for a period of 1 hour. Nonconformance to 3.7.1, 3.7.2, 3.7.3, or 3.10 shall constitute failure of this test.

4.5.2.2 Drop. The oiler shall be filled with oil conforming to MIL-L-46167, completely assembled and dropped from not less than 4 feet on a flat concrete surface. After being subjected to a temperature of -25 °F for a period of 24 hours:

a. Drop the oiler 3 times to impact upon a side.

b. Drop the oiler 3 times to impact on the bottom.

Nonconformance to 3.7.1, 3.7.2, 3.7.3, 3.9, or 3.10 shall constitute failure of this test.

4.5.2.3 Dispensing. The oiler shall be filled with oil conforming to MIL-L-46167 and subjected to a -25 °F temperature for a period of 24 hours. After conditioning, the oiler shall be held in the horizontal position and operated by hand until the flow of oil ceases. Measure the oil remaining in the oiler. Point the spout downward and operate the oiler by hand. Nonconformance to 3.7.1, 3.7.2, 3.7.3, 3.8 or 3.10 shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of packaging.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

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4.6.1.2 Sampling. Sample size shall be determined by using MIL-STD-105, tables 1 and 2. A lot shall be accepted when no defects are found and rejected when one or more defects are found.

4.6.1.3 Examination. Samples selected as specified in 4.6.1.2 shall be examined for the following defects.

113. Containers not as specified for level A or C (see 5.1.1).
114. Oilers of unlike description preserved together for level A (see 5.1.1).
115. Quantities preserved together are not as specified for level A (see 5.1.1).
116. Strapping not as specified (see 5.2.1).
117. Marking missing, illegible, incorrect, or incomplete (see 5.3).

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C as specified (see 6.2).

5.1.1 Level A. Oilers of like description shall be preserved together in quantities of 12 in a close-fitting box conforming to PPP-B-636, W6c.

5.1.2 Level C. The oilers shall be preserved as specified for level A except the boxes shall be class domestic.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. Oilers of like description preserved as specified in 5.1, shall be packed in close-fitting boxes conforming to PPP-B-601, overseas type, style optional. Box closure and strapping shall conform to the appendix to the box specification.

5.2.2 Level B. Packing shall be as specified for level A except that boxes shall be domestic type.

5.2.3 Level C. Oilers of like description, preserved as specified in 5.1, shall be packed in close-fitting wooden boxes that will prevent damage to the contents during shipment.

5.3 Marking. In addition to any special marking specified in the contract or purchase order (see 6.2), marking shall conform to MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The oilers covered by this specification are intended to dispense lubricating oils.

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6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

c. Specification part number of oiler required (see 1.3).

d. When first article is not required (see 3.2).

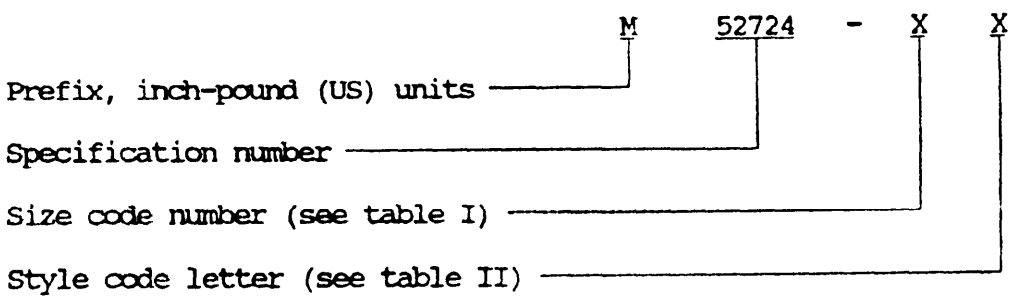
e. Timeframe required for submission of first article and number of units required (see 3.2).

f. Level of preservation and packing required (see 5.1 and 5.2).

g. Any special marking (see 5.3).

6.3 First article. When a first article inspection is required, the item(s) should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Part or Identifying system. The specification part number for the oilers covered by this specification shall be designated in the following form (see 3.12 and 6.2).



Example: M52724-1B is the definitive specification part number for 8-ounce plastic hand oilers with angle spouts (B) conforming to MIL-O-52724.

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6.4.1 Size code number. See table I.

TABLE I. Size code number.

Size Symbol	Description
1	8-ounce capacity
2	16-ounce capacity

6.4.2 Style code letter. See table II.

TABLE II. Style code letter.

Symbol	Description
A	Straight spout
B	Angle spout

6.5 Subject term (key word) listing.

Oiler, hand
Oiler, lubrication dispensing
Oiler, plastic

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - ME

Review activity:
DLA - CS

Preparing activity:
Army - ME

Project 4930-A243

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-O-52724B(ME)	1990/1/10

3. DOCUMENT TITLE
Oilers, Hand, Plastic

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
	(703) 664-5717	354-5717
c. ADDRESS (Include Zip Code) US Army Belvoir RDE Center ATTN: STRBE-TSE Ft. Belvoir, VA 22060-5606	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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